Who's Who in the Anemone? A Benchmark for Clownfish Tracking in the Wild

Owen Chau[1*] **Alexander Gou**[1*] **Ruta Sirdeshmukh**[1*] **Sarah Z. Zhou**[1*]
Michael Wakeham[1†] **Quang Tuan Truong**[1‡] **Logan Lawrence**[2‡]
Andy Zhang[1‡] **Veronika Tio**[1‡] **Jaahnavi Tirumala**[1] **Arjun Chandra**[1]
Fritz Alexander Francisco[1] **Peter Buston**[1] **Boqing Gong**[1]
[1]Boston University [2]University of Massachusetts, Amherst

Abstract

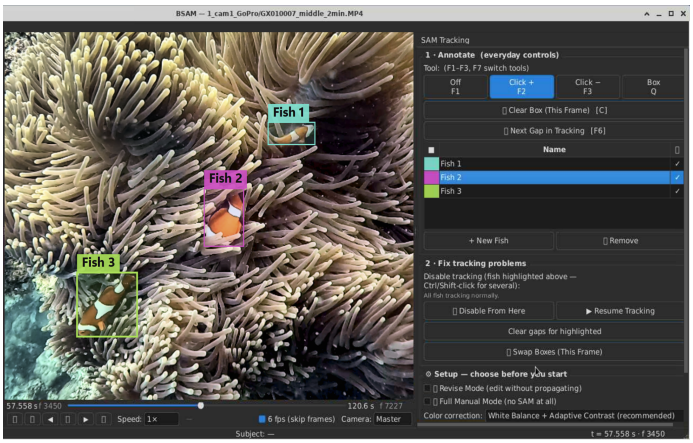
Clownfish behavior and social structure offer interesting opportunities for marine biology research, however their analysis often requires time-consuming manual annotation of video data. Although computer vision can provide a means for automating this process, existing methods still struggle in long-range settings when fish move rapidly, appear visually similar, or frequently become occluded. To assess these challenges and motivate further research in this domain, we construct a clownfish tracking dataset for training and evaluation of multi-object tracking algorithms. We facilitate dataset construction by developing a semi-automated annotation tool, which we use to annotate 30,071 bounding boxes over 12,771 frames of clownfish video. This high-quality data allows us to benchmark and fine-tune several object detection and tracking pipelines, providing insight into their performance and failure modes within this challenging domain.

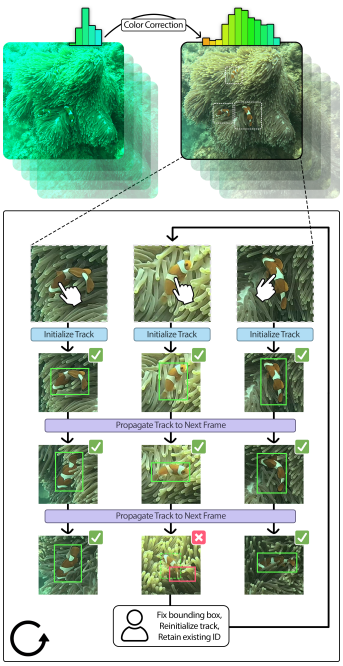
1 Introduction

Clownfish and anemone reef systems are known to be important bioindicators of climate change, as coral bleaching occurs due to the increase in ocean temperatures associated with climate change [1, 2]. This bleaching severely degrades the anemones that clownfish depend on for survival. To measure the change in coral reef systems and their effect on clownfish, biologists must first understand and track the social structure and population dynamics of clownfish over time. Doing such behavioral analysis requires laborious annotation of long-range footage, with multiple passes often needed to ensure accuracy. Passing this workload to automated pipelines offer a potential solution for scalable behavior classification. However, these algorithms require localization and tracking of fish as a prerequisite to classifying behavior. Existing object tracking algorithms such as Segment Anything Model (SAM 3) [3], You Only Look Once (YOLO) [4, 5], or GroundingDINO [6] struggle with specific failure modes that clownfish tracking data exhibit, including frequent occlusion, quick darting motions, and identity swaps with visually-similar fish. To address this challenge and evaluate the current landscape of tracking in this domain, we build a semi-automatic annotation tool and annotate a diverse set of clownfish video data. This allows us to obtain consistent long-range tracking annotations over 12,771 frames of video, totaling over 30,071 bounding boxes. From these annotations, we benchmark current tracking and identification methods, as well as substantially improve performance by fine-tuning them on our data. By supporting the development of more reliable tracking methods, our dataset provides a foundation for automatic analysis of clownfish in natural reef environments. This could help marine ecologists study how climate change, such as rising ocean temperatures, affects clownfish behavior and social structure over time.

[*]Equal contribution. [†]Corresponding author. [‡]Equal core technical contribution.

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Dataset	Frames	Boxes	Setting	Long-range	Track span (s)	Occl. gap (s)	Occl. freq (%)
FishTrack23 [7]	—	~850,000	Mixed Underwater	✗	—	—	—
FISHTRAC [8]	3,449	—	Natural Reef	✗	—	—	—
BrackishMOT [9]	16,618	49,266	Brackish water	✗	5.14	0.65	0.91
DeepSeaMOT [10]	2,400	57,376	Deep sea	✗	7.74	1.66	2.53
MFT25 [11]	48,066	408,578	Aquaculture	✓	91.83	3.08	3.53
Ours	**12,771**	**30,071**	**Reef Anemone**	✓	**114.35**	**1.47**	**16.97**

Figure 1: **Annotation workflow and fish tracking dataset comparison.** Left: our annotation interface and comparison with existing fish tracking datasets. Track span measures the average full duration of each fish track, occlusion gap measures the average duration of full occlusion for a fish, and occluded frequency measures the average percentage of time a fish is fully occluded. Right: our semi-automatic annotation workflow using SAM 3 [3].

1.1 Related Work

Fish tracking datasets. Multi object tracking for aquatic animals has been studied through broad underwater benchmarks and species specific datasets that emphasize natural background clutter, scale variation, schooling, unreliable detections, and nonlinear motion [7, 9, 10, 12, 13]. FishTrack23 [7] collects varied underwater sequences with persistent identity linked bounding boxes, while BrackishMOT [9] focuses on small schooling fish in turbid natural water. FISHTRAC [8] studies tracking under weak and noisy detections using continuous confidence information, while MFT25 [11] targets nonlinear swimming, scale changes, and frequent overlap between fish. *Our dataset differs by combining field recorded clownfish in their natural anemone with long-range persistent identities, frequent occlusion, fast motion, small targets, and repeated disappearance and reentry.* Figure 1 summarizes these datasets.

2 Dataset

2.1 Data Collection

As part of an ongoing study (Francisco & Becker et al., in prep.), we collect underwater video data of clownfish (Amphiprion percula) within their natural anemone habitat on the inshore reefs of Kimbe Bay, Papua New Guinea. These data were collected for the ongoing study and have not yet been published, building on previous research which has examined *A. percula* in the same region [14, 15, 16]. Each session is recorded using three GoPro Hero 12 cameras positioned 50-100 cm above the host anemone and angled towards it. Cameras records at 4K resolution with 16:9 aspect ratio, 60 FPS, and no automatic color correction. Although we capture multiple views, we focus the current benchmark on single view and leave multi-view tracking as future work. We select seven primary two-minute clips and 5 clips of varying length, totaling ~18 minutes of video data to be annotated. To ensure diversity, we select sessions based on host-anemone bleaching, the presence of a breeding pair, the number of resident clownfish, and recording depth. Sampling every tenth frame then produces 6 FPS clips for annotation.

2.2 Annotation

Fully manual annotation requires constructing a bounding box in every frame, including intervals where a fish remains clearly visible and its motion easily inferrable. This spends substantial human effort on easy tracking segments, rather than difficult cases where it is most needed. We therefore develop a semi-automatic annotation tool built around SAM 3 [3]. Automatic tracking propagates human-annotated localization through straightforward segments of each clip, allowing annotators to focus on more challenging cases involving occlusion, rapid motion, and identity ambiguity.

Semi-automatic workflow. Before annotation, we apply two-stage color correction: automatic white balancing removes color outliers and rescales color values, followed by brightness enhancement to improve contrast (example in Figure 1). Annotators initialize each fish with a point prompt, from which SAM 3 generates a segmentation mask and propagates a candidate trajectory through subsequent frames. Multiple fish are initialized similarly and propagated jointly. We convert the masks to axis-aligned bounding boxes, which annotators review and correct for localization errors or missing detections. The tool then exports frame-level bounding boxes with persistent trajectory identities. Figure 1 visualizes the semi-automatic workflow.

Annotation protocol. We annotate each clownfish with a tight axis-aligned bounding box around its visible extent, without estimating occluded body regions. A single bounding box therefore encloses all visible portions of a partially occluded fish, while fully occluded fish receive no annotation. Each fish is annotated separately, even when multiple fish overlap, and is assigned its own persistent trajectory identity. This identity is retained through temporary occlusion and reused if the fish leaves the camera view and later returns. Annotators distinguish returning fish using their size and stripe pattern. Frames with uncertain localization or identities are flagged for later review rather than resolved through uncertain judgment. The resulting dataset contains 12,771 frames, with 30,071 annotated bounding boxes over 37 trajectories.

Inter-annotator agreement. Each test clip is independently labeled by two annotators following the same protocol. For each tracking metric, we first treat one annotator's labels as the ground truth and the other's as predictions, then reverse the roles and average the two results. These scores now provide a reference for annotation ambiguity, used in comparison to the tracking methods we evaluate.

3 Experiments

Baselines. We benchmark four different tracking methods. SAM 3 [3] performs prompt-conditioned video segmentation and tracking, which we evaluate using text and ground-truth click initialization. GroundingDINO [6] and YOLOv8l-Worldv2 [5] perform open-vocabulary frame-level detection, with ByteTrack [17] used to associate predictions across consecutive frames. SU-T is a fish-specific tracker that combines unscented Kalman filter with FishIoU matching. We evaluate each approach both off the shelf as well as fine-tuned on our data.

Metrics. Each tracker is evaluated using standard MOT metrics [18], reported in Table 1. **HOTA** is the primary score, combining detection, association, and localization into a single accuracy. **IDF1** is the identity-preserving F1 over whole trajectories, and is the metric most sensitive to identity swaps. **MOTA** is a detection-dominated CLEAR accuracy that penalizes false positives, missed detections, and identity switches. **AssA** is HOTA's association half, which degrades under occlusion and identity swaps. **DetA** is HOTA's detection half, which degrades on small, fast, or occluded targets. **IDs** counts identity switches.

3.1 Results

Off-the-shelf performance. Overall, SAM 3 performed best when given this task zero-shot. Despite click-prompted SAM 3 receiving a ground-truth point to initialize the track, it surprisingly does consistently worse than text-prompted SAM 3 on all metrics. This result emphasizes that accurate initial localization alone does not ensure reliable long-range tracking. We also find that GroundingDINO+ByteTrack and YOLO+ByteTrack perform very poorly within this domain, while SU-T is trained within a closer setting and consequently has reasonable performance.

Table 1: **Main results.** Multiple object tracking results on our clownfish benchmark. Inter-annotator agreement is included as a reference. Bold and underline indicate the best and second-best results within each training setting, respectively. ↑: higher is better. ↓: lower is better.

Method	HOTA ↑	IDF1 ↑	MOTA ↑	AssA ↑	DetA ↑	IDs ↓
Off-the-shelf, no training on our data						
SAM 3 (text prompt) [3]	**55.7**	**65.4**	**53.9**	**59.7**	**54.1**	**19.5**
SAM 3 (GT-click prompt) [3]	<u>41.8</u>	<u>50.9</u>	<u>3.8</u>	<u>46.4</u>	<u>38.2</u>	<u>37.5</u>
GroundingDINO [6] + ByteTrack [17]	7.8	5.5	-53.8	4.0	15.3	109.0
YOLO [5] + ByteTrack [17]	2.8	1.8	-153.0	1.9	4.0	323.0
SU-T [11]	24.1	19.4	-54.7	30.4	19.5	139.0
Trained on our clownfish annotations						
SAM 3 (text prompt) [3]	26.3	29.9	-72.1	<u>36.2</u>	19.7	**15.5**
SAM 3 (GT-click prompt) [3]	**56.7**	**71.7**	**46.3**	**61.3**	**52.8**	<u>29.5</u>
GroundingDINO [6] + ByteTrack [17]	9.0	5.3	-58.7	5.1	16.4	110.0
YOLO [5] + ByteTrack [17]	20.1	19.5	<u>40.5</u>	9.2	<u>44.3</u>	333.0
SU-T [11]	<u>33.8</u>	<u>36.2</u>	-0.5	33.6	34.9	109.0
Inter-annotator agreement	78.1	87.8	77.4	84.0	73.4	8.0

Fine-tuning performance. We split the dataset by recording session, assigning each full clip to a single training, validation, or test split. This prevents closely related frames from appearing across multiple splits. To increase the amoutn of training data, we also include preliminary annotations with coarser bounding boxes. The validation and test set contain only the high-quality annotations.

Further training on our clownfish annotation affects each method differently. SAM 3 with GT-click prompting improves on every metric and becomes the strongest method. YOLO+ByteTrack and SU-T also improve across most metrics, but stay relatively the same for identity swaps. GroundingDINO changes little from this training, and in contrast to all other methods, fine-tuning text-prompted SAM 3 leads to degraded results.

Analysis. Fine-tuning has opposite effects on the two SAM 3 prompting conditions, where click-prompted performance is improved while text-prompted performance decreases. We find that training the language backbone yields performance comparable to keeping it frozen, and that convergence occurs much quicker than the click-prompted counterpart, suggesting early overfitting. This degradation in text-prompted performance also appears to arise from detection rather than identity association: DetA falls from 54.1 to 19.7 after fine-tuning, while the number of identity switches actually decreases from 19.5 to 15.5. In contrast, click-prompted SAM 3 improves in both detection and association accuracy after fine-tuning.

Even after these gains, however, performance for fine-tuned SAM 3 with click prompt remains limited in scenarios with small fish. Recall is 10.6% for instances smaller than 32 pixels, compared with 60.9% for instances larger than 96 pixels. This large recall gap suggests that the remaining performance gap is at least partly driven by difficulty detecting small targets, even after domain-specific training. Overall, these results highlight how even the strongest fine-tuned method still trails behind inter-annotator agreement, leaving considerable room for further improvement in fish tracking.

4 Conclusion

We introduce a clownfish tracking dataset and semi-automatic annotation pipeline designed to support scalable analysis of clownfish behavior in natural reef environments. The dataset contains 30,071 bounding boxes across 12,771 annotated frames, capturing long-range motion, frequent occlusion, rapid motion, and visually similar individuals. We benchmark a variety of tracking methods on our dataset, where we find text-prompted SAM 3 achieves the strongest off-the-shelf performance, while fine-tuning yields substantial but model-dependent improvements. Nevertheless, reliable tracking still remains a key barrier in converting large-scale underwater video data into behavioral analysis. Our contributions help address this challenge by providing a foundation for scalable studies of clownfish behavior and its response to climate change in reef ecosystems.

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